

Response to Reviewer 3 (electronics-4525706, Revision 2)

We thank the reviewer for the second-round assessment and for keeping the three validation requests on the table. The main change in this revision is a new real-robot experiment: the mission stack of Section 6 was run unchanged on the physical AMMR in the test room, driven from the management console with the same knowledge graph, and mirrored in the Isaac Sim twin (new Section 6.3, Table 14, Figure 20, and Supplementary Video S1). The remaining two requests are addressed by scoping and by stating precisely what independent evidence the paper does and does not contain. All section numbers refer to the revised manuscript.

Comments 1: Independent building and facility validation has not been performed; the authors have identified this as future work. The revised article lacks full end-to-end validation involving different scanning conditions, different building geometry, and an independent facility.

Response 1: We agree that a second building is not part of this study, and we have kept that statement explicit rather than weakening it. What the paper does contain, and what we have made easier to find, is the cross-floor cafeteria scene: a different floor of the building with different room geometry (open cafeteria versus equipment-filled test room), a different scanning campaign, and no overlap with the scenes on which any threshold was developed. Section 5.8 reports the frozen pipeline end-to-end on that scene (segmentation, verification tiers, and the held-out structure filter, 2/2 remnant clusters flagged with zero false positives) and Section 7 ("Scope") states the bound it provides. The new real-robot experiment (Section 6.3) adds a second kind of independence: the robot's own localization map is built independently of the scan, and the knowledge graph produced by the framework is consumed by an external navigation stack through a single calibrated frame offset. We did not add a new building within the revision window because a facility annotated to a lower ground-truth standard than the rest of the paper would weaken rather than strengthen the evidence; multi-site validation remains stated as future work in Sections 7 and 8.

Comments 2: There is no end-to-end task validation on a real robot.

Response 2: This is now addressed with hardware. Section 6.3 ("Real-Robot Mission Execution") reports semantic missions executed on the physical AMMR in the test room: the management console issues the command, the mediator resolves and gates it against the knowledge graph, the behavior tree executes it through the robot's on-board Nav2 stack, and the Isaac Sim twin mirrors the robot from the same store. Three semantic drives completed (two `goto_object` tv approaches and one `goto_place` drive into the monitor-workstation region, 3.6–4.0 m each) in 14–22 s with arrival errors of 0.22–0.34 m (mean 0.30 m); a place command issued while the robot already stood inside the region completed without motion; and the owner-refuted phantom `keyboard` was refused by the gate in under one second with zero motion, twice, exactly as in simulation. Table 14 lists every run, Figure 20 shows two filmed missions frame by frame (robot, twin, console), and Video S1 provides the synchronized footage. We state the scope honestly in Sections 6.3 and 7: one room, one platform, three completed drives, each issued from a standing start; a longer multi-mission, multi-site campaign is future work. The abstract, introduction, and conclusion now mention the real-robot result.

Comments 3: The requirement for independent ground truth has not been fully met.

Response 3: We accept that the ground truth is not fully independent and have not claimed otherwise. The paper now carries four layers of evidence on this point, each stated with its limits: (i) the owner audit, whose anchoring channel is disclosed (Section 5.1); (ii) a blinded second annotation by a co-author who had seen neither the pipeline output nor the owner audit, agreeing with the owner on 31/33 clusters and reproducing 25/26 of the owner's overrides of the model (Section 5.1), which argues against those corrections being driven by anchoring; (iii) the parameter-sensitivity re-scoring of Section 5.2, which grades the segmentation sweep against the owner-audited inventory in a membership-independent form (all clean-cloud points inside each audited box), so that no clustering decision defines the reference; and (iv) the new external blind annotation described below. For this

revision we additionally obtained a blind annotation from an external annotator with no connection to the work and no familiarity with the facility (same 41 sheets, same protocol; Section 5.1). Without site knowledge, agreement with the owner is 21/33 (63.6%), but 16/19 (84%) of the labels given at confidence 4–5 agree, and 11 of the 12 disagreements fall on absorbed/merged industrial objects and ceiling/wall remnants — the categories the paper identifies as render-ambiguous and as the reason on-site owner knowledge is needed. We report this as the honest bound on render-only independent annotation rather than as a replacement for the owner audit: distinctive furniture and robots are labeled consistently by anyone, whereas the objects that required physical inspection in the owner audit are exactly those an unfamiliar annotator cannot resolve from renders. The co-author annotation remains disclosed as such in Section 5.1. The real-robot experiment adds an independent functional check of the verified graph: the mission targets the robot reached were verified nodes, and the goal it refused was the owner-refuted phantom.